Armlab: Vision-Guided Autonomous Block Manipulation with a 6-DOF Lite 6 Arm

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***Abstract*—We present an autonomous pick-and-place system for a UFACTORY Lite 6 arm with an overhead Intel RealSense L515 RGB-D camera. Our Denavit–Hartenberg and product-of-exponentials forward kinematics agree with each other to 10^{-11} mm and with the vendor to 2.1 mm on average. A closed-form inverse kinematics solver with an explicit branch and singularity policy round-trips all 500 test poses to floating-point precision, and a damped least-squares numerical solver does the same on the 457 it solves. The camera is calibrated from four AprilTags (0.3 px reprojection RMS), with depth–color registration and a per-pixel depth-bias correction. The block detector works on height rather than color and rejects distractors with nine geometric and photometric tests; it found 50 of 52 test blocks with a mean position error of 4.4 mm. A plan-before-moving pick-and-place layer runs the four competition events, sorting, unstacking and stacking blocks from previously unseen layouts with distractors.**

I. Introduction

A manipulator must know where objects are and which joint angles put the gripper there. This lab builds both from first principles on a UFACTORY Lite 6 (6-DOF, spherical wrist, 440 mm reach) with a two-finger Gripper Lite and a ceiling-mounted Intel RealSense L515 above a 50 mm-gridded board [1]. The objects are 25 mm ("S") and 38 mm ("M") cubes in six colors.

The system (Fig. 1) has a *kinematics and control* half that maps world poses to joint motions, and a *vision* half that maps camera pixels to world points and finds, sizes, orients and colors the blocks. Our contributions are verified DH and PoX forward kinematics; a closed-form IK solver with a deterministic branch policy, cross-checked numerically; AprilTag extrinsics with depth registration and bias correction; a height-based block detector with explicit distractor rejection; and an event layer that unstacks, sorts, lines up and stacks blocks.

II. Methodology

All quantities use the board frame: origin at the center of the arm's mount, $z = 0$ on the board surface, millimeters and radians.

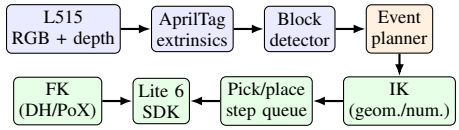


Fig. 1. System overview. Blue: vision (pixels → world). Green: kinematics and control (world → joints). Orange: task-level decisions.

TABLE I
Standard DH table of the Lite 6 (flange frame).

Joint	θ_i^0 (rad)	d_i (mm)	α_i (rad)	a_i (mm)
1	0	243.3	$-\pi/2$	0
2	$-\pi/2$	0	π	200
3	$-\pi/2$	0	$\pi/2$	87
4	0	227.6	$\pi/2$	0
5	0	0	$-\pi/2$	0
6	0	61.5	0	0

A. Kinematics and Control

1) Teach-and-repeat: In hand-guiding mode, a *Record* button stores the six joint angles and the gripper state as one waypoint, so a grasp is two consecutive waypoints with the same pose and different gripper states, and the two cannot drift apart. Playback checks every waypoint against the joint limits, moves with `wait=True`, then actuates the gripper with a 0.8 s dwell (the Gripper Lite has no feedback). We taught a 25-waypoint, three-block stack (Appendix, Table XI).

2) Forward kinematics (DH): Table I gives our standard DH parameters [2]; the frames are in Fig. 2a. Each row gives

$$^{i-1}T_i = \mathrm{Rot}_z(\theta_i{+}\theta_i^0)\,\mathrm{Trans}_z(d_i)\,\mathrm{Trans}_x(a_i)\,\mathrm{Rot}_x(\alpha_i), \tag{1}$$

and $^0T_6 = \prod_i {}^{i-1}T_i$ is the flange pose. Orientation is roll–pitch–yaw about fixed axes, $R = R_z(\psi)R_y(\theta)R_x(\phi)$, as in the SDK. The zero pose is $(87, 0, 154.2)$ mm.

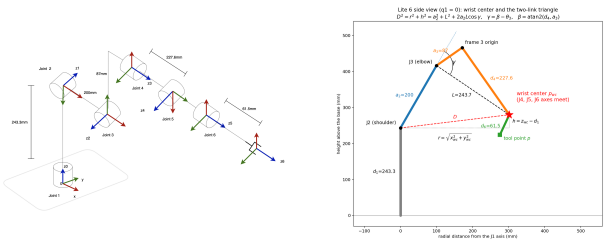


Fig. 2. (a) DH frames (red x, green y, blue z) with link lengths; all six screw axes are the joint z-axes at zero. (b) Side view in the plane of q_1: the wrist center p_{wc} (star), one d_6 back from the tool point, and the two-link triangle (a_2, L, D) that sets q_2 and q_3.

3) Forward kinematics (PoX): With every joint at zero, the flange pose is

$$M = \begin{bmatrix} 1 & 0 & 0 & 87 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 154.2 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{2}$$

Each space-frame screw axis is $\mathcal{S}_i = [\omega_i; v_i]$ with $v_i = -\omega_i \times p_i$ for any point p_i on the joint axis [3]; a second point on each axis gave the same v_i. One axis per column,

$$\mathcal{S} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 1 & 0 \\ 1 & 0 & 0 & -1 & 0 & -1 \\ 0 & -243.3 & 443.3 & 0 & -215.7 & 0 \\ 0 & 0 & 0 & 87 & 0 & 87 \\ 0 & 0 & 0 & 0 & 87 & 0 \end{bmatrix}, \tag{3}$$

and $T(q) = e^{[\mathcal{S}_1]q_1} \cdots e^{[\mathcal{S}_6]q_6} M$. M was checked against $\mathrm{FK}_{\mathrm{DH}}(0)$, so the vendor comparison is the independent check of both forms.

4) FK verification: For 500 joint vectors drawn uniformly within the limits, we compared our pose with the vendor's `get_forward_kinematics`. Position error is the Euclidean distance; orientation error is the angle of $R_{\text{ours}}^{\top} R_{\text{SDK}}$, which avoids Euler-angle wrap-around. The firmware's own table (`arm.dh_params`) shifts α and a by one row relative to standard DH, so chaining it with our code misses the vendor by 400–500 mm; it could not serve as a control.

5) Geometric IK: A top-down grasp places the flange 90 mm above the block center (Section IV-A explains why not the handout's 85 mm). The spherical wrist decouples position and orientation [4] (Fig. 2b; 3D view in Appendix Fig. 13):

$$p_{wc} = p - d_6\, R\, \hat{z}, \qquad q_1 = \operatorname{atan2}(y_{wc}, x_{wc}), \tag{4}$$

$$r = \sqrt{x_{wc}^2 + y_{wc}^2}, \qquad h = z_{wc} - d_1, \tag{5}$$

$$D^2 = r^2 + h^2. \tag{6}$$

The elbow offset a_3 and forearm d_4 form a fixed right angle, which we fold into one link $L = \sqrt{a_3^2 + d_4^2} = 243.7$ mm at the fixed angle $\beta = \operatorname{atan2}(d_4, a_3)$. The law of cosines then gives

$$\cos\gamma = \frac{D^2 - a_2^2 - L^2}{2 a_2 L}. \tag{7}$$

If $|\cos\gamma| > 1$ (wrist center farther than $a_2 + L = 443.7$ mm or closer than $L - a_2 = 43.7$ mm from the shoulder), or no branch is inside the joint limits, the solver returns `None` and callers never move. The elbow branches are $\theta_3 = \beta \mp \gamma$, and with $g = \beta - \theta_3$,

$$\theta_2 = \operatorname{atan2}(-h, r) - \operatorname{atan2}(L \sin g,\ a_2 + L\cos g), \tag{8}$$

where $-h$ appears because frame 1's y-axis points down. Removing the DH offsets gives $q_2 = \theta_2 + \pi/2$ and $q_3 = \theta_3 + \pi/2$. For the wrist, $R_6^3 = (R_3^0)^{\top} R = [r'_{ij}]$ gives

$$q_5 = \operatorname{atan2}\big(\pm\sqrt{r'^2_{13} + r'^2_{23}},\ r'_{33}\big), \tag{9}$$

$$q_4 = \operatorname{atan2}(\mp r'_{23}, \mp r'_{13}), \tag{10}$$

$$q_6 = \operatorname{atan2}(\mp r'_{32}, \pm r'_{31}). \tag{11}$$

Branch policy. The solver returns the solution nearest a reference q_{ref} (the current angles during a task, else Q_{DEFAULT}), so the same pose always gives the same answer. It (1) discards solutions outside the joint limits; (2) prefers q_1 facing the target over $q_1 + \pi$, so the base never swings round; (3) prefers the Q_{DEFAULT} elbow branch ($q_3 \in [-20.9°, 159.1°]$), using the other only when every solution on the first violates the J2 or J5 limit; and (4) takes the wrist solution ($\pm q_5$) nearest q_{ref}. A fixed sign of q_5 made J4 and J6 jump $180°$ whenever a path crossed $q_5 = 0$. J1, J4 and J6 travel $\pm 360°$, so each takes the $2\pi k$ turn nearest the reference.

Singularities. When $|s_5| < 10^{-6}$ only $q_4 + q_6$ is determined: J4 holds its reference value and J6 takes the rest ($q_5 = 180°$ is outside J5's $\pm 124°$ limit). With the wrist center on the J1 axis, q_1 is set to 0, then moved by whole turns toward q_{ref}.

6) Numerical IK: The numerical solver minimizes $e(q) = [\,p(q) - p^*;\ \ w_{\text{rot}}\,\mathrm{rotvec}(R(q)R^{*\top})\,]$ with $w_{\text{rot}} = 200$ mm/rad, using a finite-difference Jacobian and Levenberg–Marquardt steps [5]:

$$\big(J^{\top}J + \lambda\,\mathrm{diag}(J^{\top}J)\big)\Delta q = -J^{\top}e, \tag{12}$$

with λ divided or multiplied by 10 after a step that lowers or raises the cost. Joints pushed into a limit are frozen and the step re-solved; steps are capped at 0.3 rad. It reports success only if $\|e\| < 10^{-3}$, so converging to the nearest reachable pose counts as unreachable. It is seeded from Q_{DEFAULT} for reproducibility.

7) Pick-and-place motion plan: Every pick and place is *solved in full before the arm moves*; if any pose has no IK solution, nothing moves and the status bar says why. The grasp flange height is $z_g = z_c + 90$ mm for a block

centered at z_c, raised if needed so the fingertips (100 mm below the flange) stay 3 mm above what the block stands on (on a known stack, from nominal block heights, since measured tops read a few mm low). The travel height $z_t = z_{\text{tallest}} + \max(100,\, 90 + s/2) + 40$ mm keeps the fingertips or held block 40 mm above everything on the board. A pick opens, moves above the block, descends vertically, closes and lifts along the same line; a place mirrors it, releasing 3 mm above the surface, so nothing moves laterally at grasp height.

A joint-space move along a vertical line bows sideways by up to 16 mm over 100 mm, so we check it with FK at 19 points and split it at the midpoint until the drift is below 1.5 mm, then send the pieces as one blended path (`MoveArcJoint`, 10 mm radius). Each grasp is solved at the four yaws a cube allows, keeping the solution that minimizes $\sum |q - q_{\text{ref}}| + |q_6|$; the $|q_6|$ term stops J6 winding toward its limit. In click mode, a click within 30 mm of a detected block snaps to its measured center, size and yaw.

Failing safely. The plan runs one step per 20 Hz state-machine tick, so *STOP* halts the arm after the current move. A move that fails with the gripper closed stops the arm without opening it, so a held block is never dropped; any other failure stops click mode with the error shown, and the events look again, replan, and stop after three failures in a row. The gripper has no sensor, so "holding" only means the last command was *close*.

B. Camera Calibration and Pixel-to-World

1) Intrinsics: We used OpenCV's [6] checkerboard calibration [7] (8×6 corners, 25 mm squares, 10 frames) in three runs: (A) board flat and slid around, (B) board tilted through the whole volume, and (C) a repeat of B.

2) Hand-measured extrinsics: We measured the camera's optical center with a tape at $(500, 25, 942)$ mm and modeled the rotation as $R = R_y(199°)R_z(270°)$, a 19° tilt estimated with a phone's IMU (Fig. 3):

$$ {}^{w}T_{c}^{\text{hand}} = \begin{bmatrix} 0 & -0.9455 & -0.3256 & 500 \\ -1 & 0 & 0 & 25 \\ 0 & 0.3256 & -0.9455 & 942 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{13} $$

We tested it by projecting four board points, $\tilde{u} = K\,({}^{w}T_c)^{-1}X_w$, against pixels read off the live video.

3) AprilTag extrinsics: Four `tagStandard41h12` tags [8] lie at $(100, \pm 300, 0)$ and $(400, \pm 300, 0)$ mm. *Calibrate* matches the detected tag centers to these points with `cv2.solvePnP` (iterative, homography-seeded) and the factory K, stores $[R|t]$ and its inverse, and prints the camera position $-R^\top t$ and the tag reprojection RMS as checks.

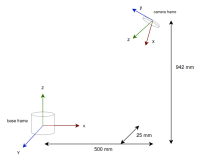


Fig. 3. Base and camera frames with the hand-measured offsets.

4) Depth registration and bias: The L515's depth–color alignment was off by up to 10 px, most at the left of the image, independent of our filters. Measured on the edges of 235 blocks in 11 snapshots, we model it as $\Delta u = 5.26 - 0.0143(u - c_x)$, $\Delta v = -0.70 - 0.0173(v - c_y)$ px and resample every depth frame. The range also has a bias. At *Calibrate*, the extrinsics give the true range along every table ray, and we fit a quadratic surface $b(u, v)$ to measured minus true range, with a median gate and 3σ clipping to reject blocks and the arm.

5) Pixel to world: For pixel (u, v) with raw depth d and depth scale s_d:

$$ Z_c = s_d\, d(u, v) - b(u, v), \tag{14} $$

$$ X_c = \frac{(u - c_x)Z_c}{f_x}, \quad Y_c = \frac{(v - c_y)Z_c}{f_y}, \tag{15} $$

$$ [X_w, Y_w, Z_w, 1]^\top = {}^{w}T_c\,[X_c, Y_c, Z_c, 1]^\top. \tag{16} $$

A zero depth returns `None`. The calibration checks (Section III-B) set $b = 0$, or they would pass by construction.

6) Top-down homography: The rectangle $x \in [-100, 500]$, $y \in [-490, 490]$ mm maps by a homography [9] to a 1372×840 px view at 1.4 px/mm, world $+x$ up and $+y$ left. We computed H with `cv2.findHomography` from the four point pairs in Table II (with four pairs the fit is exact). The source points are the rectangle's corners ($z = 0$) projected into the image through K and the AprilTag extrinsics; the destination points are the view's corners. Using projected corners rather than live tag centers keeps the view free of tag jitter. For the AprilTag calibration of (18),

$$ H = \begin{bmatrix} 1.294 & -0.2219 & -221.3 \\ 0.01899 & 1.228 & -72.90 \\ -1.420{\times}10^{-5} & -3.179{\times}10^{-4} & 1 \end{bmatrix}, \tag{17} $$

mapping camera pixels to view pixels. Clicks pass back through H^{-1} before `image_to_world`.

C. Block Detection

The detector finds blocks by *height*, not color, at about 10 Hz. Shadows, glare and the printed grid change the color image but not the height, and several distractors are block-colored, so candidates come from depth and

TABLE II
POINT PAIRS USED TO COMPUTE H.

World (x, y) (mm)	Camera pixel (src)	View pixel (dst)
$(500, \ 490)$	(180.7, 56.6)	(0, 0)
$(500, -490)$	(1206.3, 40.7)	(1372, 0)
$(-100, -490)$	(1117.1, 587.4)	(1372, 840)
$(-100, \ 490)$	(274.7, 604.8)	(0, 840)

color is read last. Every rejected region is drawn on the GUI with the failed test's name, which is how we tuned it (Fig. 10; all stages in Fig. 4).

1: Warp the registered depth into the top-down view with nearest-neighbor interpolation, which does not invent depths across block edges.
2: Fit the table plane $z = au + bv + c$ to the median depth at the four tags. Cache each tag's depth and treat a reading more than 6 mm off as a covered tag.
3: Height $= |n \cdot \mathbf{d}|\,(z_{\text{table}} - z)$, with $\mathbf{d}$ the pixel ray and n the board normal, converts range along the ray to vertical height.
4: Threshold to 15–124 mm (162 mm in Event 3), remove pixels whose 3D point lies over the 150×150 mm base (plus a 5 mm margin), open (5 px) and take external contours.
5: Keep each region's *top face*, the pixels within 5 mm of its 90th-percentile height, which drops the visible sides.
6: Run the rejection tests (Table III).
7: Project the corners of the top face's `minAreaRect` onto the top-face plane for side length (S or M within 7 mm) and yaw (mod $90°$); the center goes through H^{-1} and `image_to_world`.
8: Find the stack of up to four S/M blocks, in any order, whose height matches the top within $[-6.8, +4]$ mm; a wider block under a narrower one must show a ring, which separates S-on-M from a 57 mm cylinder.
9: Classify the median hue of the eroded top face (OpenCV units): red ≥ 165 or < 2, orange 2–18, yellow 18–35, green 35–85, blue 85–119, violet 119–165. Faces with $S < 70$ or $V < 40$ are "unknown" unless their hue is tight and their chroma $SV/255 \geq 8$.

D. Competition Strategy and Outcome

All events share one loop: *look* (merge five detector runs from a home pose that keeps the arm out of view), *decide*, *execute* one plan step per tick, and update a board model; the run looks again only when the model can no longer be trusted. The camera sees only the top of a stack, so Events 1 and 3 first unstack onto free staging spots (a 30 mm grid checked clear in the height map).

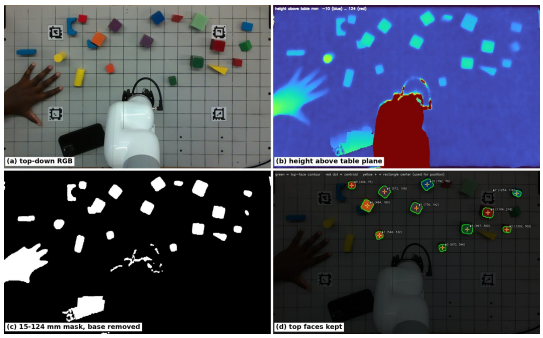


Fig. 4. Detector stages on the distractor board of Fig. 10: (a) RGB warped top-down; (b) height above the fitted table plane; (c) pixels 15–124 mm high, arm base removed; (d) top faces kept for measurement.

TABLE III
DISTRACTOR REJECTION TESTS. "REAL BLOCKS" IS THE WORST VALUE AMONG REAL BLOCKS IN OUR SAVED SNAPSHOTS, OR THE PERCENTILE MARKED (WORST: 1.32 AND 7.5 MM, BEYOND THE THRESHOLD).

Test	Reject if	Real blocks	Catches
Area	top $< 0.4A_S$ or $> 1.6A_M$	–	noise, touching blocks
Outline / top	> 2.4 (scaled for stacks)	2.06	hand 4.0, phone 3.1
Flat top	height std > 2.8 mm	2.3 mm	hand 3.3 mm
Level top	plane tilt $> 10°$	6.8°	phone 15°, bar 14°
Ramp ring	$> 1.5\times$ top area	1.2	wedges 2.0–2.2
Shape	aspect > 1.25 or fill < 0.75	1.20 (p99)	arches, prisms, bars
Size	a side > 7 mm from 25/38	4.6 mm (p95)	sponge 48 mm
Height	no S/M stack within $[-6.8, 4]$ mm	-6.7 / $+2.2$	cylinder $+7$, thin pieces -7 to -8.6
One color	circular hue std > 15	6.8	phone 26

Event 1 then builds each size's stack in rainbow order. In Event 2 every (size, color) pair has its own line slot, so blocks go in any order. In Event 3 colors within each size are assigned by *hue rank*, so one misread hue cannot put a tower out of order. Event 4 builds a main and a side tower, top-down and then with a level tool, and finally carries the side tower onto the main one, turning only J1 at 1–3% speed.

Outcome. Click-to-grab and click-to-place also worked on the arm, including stacking onto a detected block and refusing unreachable clicks without moving (videos in the Appendix). On competition day the system ran every event at level 3, autonomously and with distractors, for 2244 points (third in the AM section): Event 1 scored 300/300 and Event 4 stacked 24 blocks, both on the first try; Event 3 reached 500/500 on the third try; Event 2 scored 484/500 on the second

TABLE IV
FK VERIFICATION, 500 RANDOM CONFIGURATIONS.

Comparison	mean	median	p95	max
Position error (mm)				
DH vs. vendor	2.10	2.06	3.46	4.68
PoX vs. vendor	2.08	2.00	3.48	5.02
PoX vs. DH	$< 10^{-11}$	$< 10^{-11}$	$< 10^{-11}$	$3.5{\times}10^{-11}$
Orientation error (deg)				
DH vs. vendor	0.210	0.237	0.325	0.332
PoX vs. vendor	0.209	0.236	0.324	0.331
PoX vs. DH	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	$2{\times}10^{-6}$

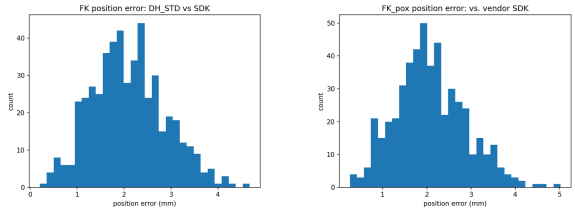


Fig. 5. Position error against the vendor: DH (left) and PoX (right).

TABLE V
IK VERIFICATION, 500 RANDOM CONFIGURATIONS. THE VENDOR ROW IS A SEPARATE HARDWARE RUN.

Solver	solved	pos. err (mm) mean/max	ori. err (°) mean/max
Geometric	500	$7.8{\times}10^{-14}$ / $2.1{\times}10^{-13}$	$1.3{\times}10^{-14}$ / $2.4{\times}10^{-13}$
Numerical	457	$6.3{\times}10^{-11}$ / $7.0{\times}10^{-10}$	$1.3{\times}10^{-11}$ / $2.4{\times}10^{-10}$
Vendor	495	2.07 / 4.74	0.180 / 0.332

try, with all twelve blocks in their slots and the lost points in line quality (3 of 5 for each size).

III. RESULTS

A. Kinematics

Table IV and Fig. 5 summarize FK accuracy over 500 random configurations; the error shows no trend with any joint angle (Appendix, Figs. 11 and 12).

Table V gives the IK round trip (FK $\rightarrow$ IK $\rightarrow$ FK) and the vendor's IK evaluated through our FK. Wherever they converge, both of our solvers recover the pose to floating-point precision, and the geometric solver converged on every configuration. Along vertical descents, radial sweeps and yaw sweeps, no joint changed by more than 5° between neighboring targets, so the branch policy produced no flips. The vendor's solutions, evaluated through our FK, land inside the 10 mm allowance.

For the reachability map (Fig. 6) we solved a top-down grasp of a 38 mm block on a 10 mm grid (6161 points) at yaw 0° and 90°, with the handout's 85 mm flange offset, and checked each solution for collisions in MuJoCo. At yaw 0°, 3488 points (56.6%) were pickable; of the other

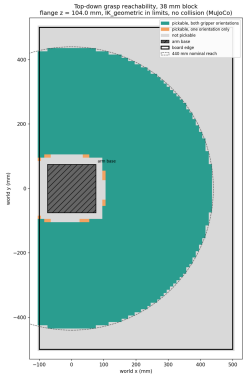


Fig. 6. Top-down grasp reachability for a 38 mm block (green: both gripper yaws; orange: one yaw). The dashed circle is the 440 mm nominal reach.

TABLE VI
INTRINSIC CALIBRATION (PX) AGAINST THE FACTORY VALUES.

	Factory	A (flat)	B (3D)	C
f_x	900.19	1283.71	901.87	902.83
f_y	900.52	1372.10	902.56	906.03
c_x	661.89	663.11	653.94	672.46
c_y	367.95	353.39	354.86	384.43
RMS	–	0.101	0.358	0.287

2673, 2259 were beyond reach, 361 on the base and 53 had the arm hit the base (Appendix, Fig. 14). 3483 were pickable at both yaws and 16 at only one. A 25 mm block changed only 29 points (0.5%), so we show the 38 mm map.

B. Calibration

Intrinsics (Table VI). Run A, with the board flat, has the *lowest* reprojection error, yet its focal lengths are 43–52% off. Runs B and C match the factory focal lengths within 0.7%, so we use the factory K.

Extrinsics. The AprilTag solve gave

$$ {}^{w}T_{c}^{\mathrm{PnP}} = \begin{bmatrix} -0.0149 & -0.9501 & -0.3116 & 478.7 \\ -0.9998 & 0.0185 & -0.0086 & 45.6 \\ 0.0139 & 0.3114 & -0.9502 & 911.4 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (18) $$

1.44° and 42.6 mm from the hand-measured matrix, with an 18.2° tilt against the hand estimate of 19°. We re-run *Calibrate* every session: the tag reprojection RMS stayed at 0.27–0.31 px (max < 0.5 px), but the camera position moved 28 mm (mostly in y) between the session of (18), H and Table II and the Oct. 15 session of Fig. 7. Table VII tests the hand-measured matrix on four board points.

Board accuracy. The projected 50 mm grid stays on the printed grid lines across the board, includ-

TABLE VII
HAND-MEASURED EXTRINSICS: PREDICTED VS. MEASURED PIXELS.

World (mm)	Predicted (px)	Measured (px)	Δ (px)
$(0, 200, 0)$	(512.4, 509.9)	(517, 518)	(4.6, 8.1)
$(0, -200, 0)$	(854.2, 509.9)	(870, 513)	(15.8, 3.1)
$(200, 200, 0)$	(502.5, 347.0)	(504, 350)	(1.5, 3.0)
$(200, -200, 0)$	(866.8, 347.0)	(879, 345)	(12.2, -2.0)

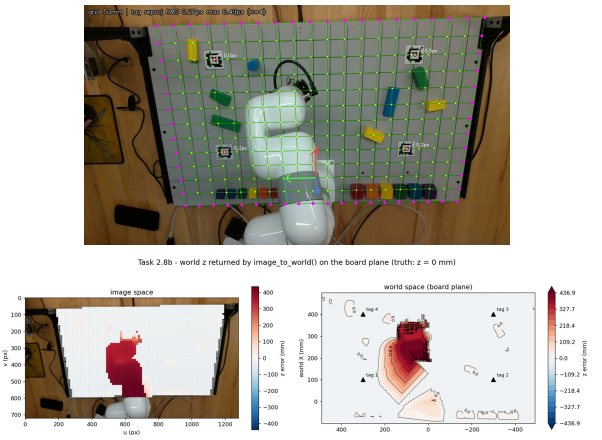


Fig. 7. Top: board grid points (yellow) projected through K and the extrinsics; red circles are the detected tag centers. Bottom: world z returned on the empty board (truth 0), bias correction off; the red blob is the arm occluding the board (Oct. 15 session).

TABLE VIII
3D ERROR (MM) BY REGION AND HEIGHT, 12 FIXTURE POSES.

Group	mean $\|e\|$	max $\|e\|$	mean (e_x, e_y, e_z)
All	15.0	21.5	$(+8.3, -1.0, -10.8)$
Center (A, C)	13.4	15.4	$(+8.4, +0.8, -9.9)$
Edge (B, D)	16.6	21.5	$(+8.2, -2.8, -11.7)$
Low (z≈70)	15.4	18.0	$(+10.3, -0.6, -10.0)$
Mid (z≈170)	14.5	18.2	$(+8.4, -1.1, -10.3)$
High (z≈270)	15.1	21.5	$(+6.2, -1.3, -12.0)$

ing the corners far from the tags (Fig. 7). Sampling `image_to_world` every 12 px over the empty board with the bias correction off (Fig. 7), 97% of samples had valid depth, the median z was -12.0 mm and the RMS outside the tag rectangle was 16.5 mm. Inside it the arm stands on the board, so that RMS (227 mm) measures the arm (Appendix Fig. 15).

3D accuracy (Table VIII). The arm held a tagged fixture with the tag $(+64, 0, +21)$ mm from the flange, an offset we measured by hand; truth was the SDK pose plus this offset, and the estimate the tag center from `image_to_world`, at four locations and three heights (Appendix, Table XII).

TABLE IX
DETECTOR RESULTS ON THE 52-BLOCK TEST SNAPSHOT.

Quantity	Value
Blocks found / placed	50 / 52 (0 false pos.)
Position error mean / median / max	4.4 / 4.3 / 8.3 mm
Side length, S (17) / M (33)	26.1 ± 0.9 / 37.3 ± 1.2 mm
Top-face height, S / M	22.1 / 36.6 mm
Color correct, Task 3.5 / final rules	43 / 49 of 52
Table-plane misfit (RMS)	0.54 mm
Depth bias (fit RMS 0.8 mm)	$+1.8$ to $+28.8$ mm

C. Block Detection

We tested the detector on a saved snapshot of a 52-block board, six rows by nine columns on the 50 mm grid (two positions under the arm base), at mixed yaws (Fig. 8, Table IX). Off-block pixels had a median height of $+0.8$ mm, well clear of the 15 mm threshold.

There were no false positives. Position error against the nearest grid intersection (Fig. 9) did not grow with distance from the board center ($r = 0.16$); the signed mean was $(-2.4, +0.2)$ mm, with blocks at $x \geq 150$ mm sitting 3–4 mm short in x. Both missed blocks (a red S and a violet S) were in the height mask but failed the shape test with a fill just under the 0.75 limit. The S and M sides are separated by a 7 mm gap, and top faces read 1.4–2.9 mm below nominal.

Color. With the hue bounds used for Task 3.5, 43 of the 52 blocks got the right color (Fig. 9): the two missed blocks, four violets read as blue, two orange S as red and one dark violet ($V < 40$) as "unknown". Two later changes, both made from event run logs (Table X; Section II-C, step 9), affect these blocks. Accepting dark faces of one clear hue recovers the dark violet (44/52); moving the red/orange boundary from 6 to 2 and the blue/violet boundary from 128 to 119 fixes both orange S ($H = 4$–6) and three violets ($H = 121$–122), giving 49/52. These counts re-score the hues logged in this snapshot; we did not rerun the detector. One violet ($H = 114$) still reads blue.

Distractors. Fig. 10 shows the detector on a board with a hand, a phone, a sponge, standing and lying cylinders, an arch, a bar and prisms among real blocks; each distractor is rejected and labeled with the test it failed. The one exception in this capture, a thin piece accepted as an S, came from the earlier height window and is rejected by the final one (Table X). Both wedges in a later snapshot fail the ramp test. Most margins in Table III are wide (Appendix Fig. 16), but the lower height bound sits 0.1 mm beyond the worst real block, and the shape and size limits cut into the real-block tail (an edge S at aspect 1.32, the two missed S at fill < 0.75): looser limits let a wedge's 20×28 mm sloped

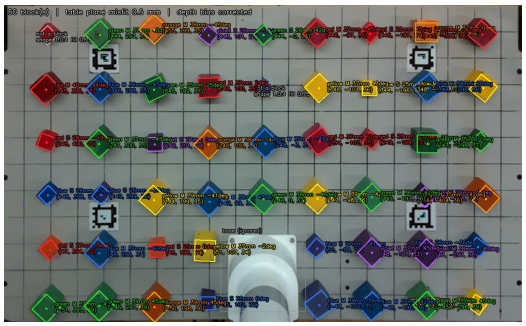


Fig. 8. Detector output on the 52-block test board: color, size, side length, yaw and world (x, y, z_{top}) for each of the 50 detections. The two rejected regions ("not a block") are the missed S blocks; the arm base is ignored.

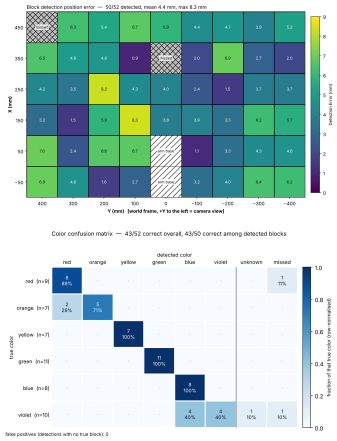


Fig. 9. Top: detector position error over the board. Bottom: color confusion matrix with the hue bounds used for Task 3.5 (43/52 correct); Section III-C and Table IX give the counts under the final rules.

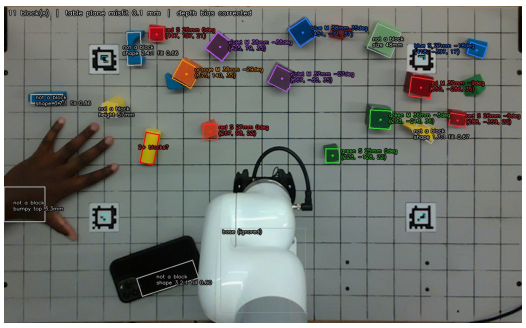


Fig. 10. Detector output on a board with distractors (Oct. 15 detector). Colored boxes: accepted blocks. Gray boxes: rejected regions with the failed test (hand: flat top; phone, bar, arch, prism: shape; sponge: size; standing cylinder: height); the lying cylinder is flagged "2+ blocks?". The thin blue piece at top right is rejected by the final height window.

top pass as an S.

IV. Discussion

A. Kinematics

The roughly 2 mm FK residual against the vendor is smooth: it does not grow with any joint, jump, or change sign. By the handout's diagnostic patterns that rules out a wrong sign, offset or α, so the residual most likely comes from the per-arm factory calibration the firmware applies and the published table omits. That floor (Table IV) is about half the detector's position error (Table IX), so FK does not limit pick-and-place. PoX and DH agree to 10^{-11} mm, but since M came from $\text{FK}_{\text{DH}}(0)$ this validates the screw axes and chaining rather than the table.

The geometric solver never failed inside the limits. The numerical solver fails on 9% of poses, which is why it is only a fallback: all 43 failures were poses the geometric solver solved, i.e. local minima of a solver seeded at Q_{DEFAULT}, typically with the target on another J1 or elbow branch; we did not test other seeds. In early runs a flange 85 mm above an S block's center put the fingertips 2.5 mm into the table, so we moved to 90 mm with a 3 mm fingertip floor. The reachability map's hole around the base and circular outer edge show that reach and the base, not the J5 limit, bound top-down grasps.

B. Calibration

Why run A is invalid. A board slid flat always presents the same orientation, so the solver cannot separate focal length from distance; it trades one against the other and still fits the corners almost perfectly. A low reprojection error therefore does not show that a calibration is good; the board has to be tilted through varied views to pin down the focal length.

Hand-measured extrinsics. The error of the hand-measured matrix (about 3–18 mm at 1 m) was largest on the $-y$ side, consistent with small errors in yaw and in the 25 mm y offset. A tape cannot locate the optical center inside the housing, and 1° of tilt moves a board point by about 17 mm, so this method is a sanity check, not a calibration for grasping.

Where the 3D error comes from. Along each viewing ray, the fixture error averages $+8.9$ mm (4–19 mm), matching the -12 mm median on the empty board and the fitted bias surface, which grows from the far to the near edge (Table IX); since that surface is fitted against our own extrinsics, part of it may be an error in the camera's distance rather than in the L515's range. Across the ray, the error component perpendicular to it is a nearly constant $+11$ mm in x (8.6–14.7 mm) at all twelve poses (unlike the raw e_x of Table XII), which an overlong range cannot cause (every ray points toward $-x$). We attribute most of it to the hand-measured fixture offset or the flange origin, which shift every pose equally

because the tool orientation was constant; its 1.6–3 mm shrinkage from the lowest to the highest pose points to a small extrinsics contribution. Error grew at the board edges (oblique view, noisier depth at grazing incidence, a pose fitted to four tags in one region); height had little effect.

C. Detection

Working on height made the detector largely insensitive to lighting (scaling the color image to $0.6\times$ and $0.4\times$ brightness did not change the accepted blocks on our snapshots) and gave stack levels for free. Its main cost was the L515's 10 px depth–color misregistration, which we had to model. The 4.4 mm position error combines the residual registration (about 2 px), depth noise on block tops and our hand placement of the blocks; the -3 to -4 mm x bias beyond $x = 150$ mm is the size of that placement tolerance, so we could not attribute it to one cause; it is well inside the gripper's tolerance.

Color errors fell where hue classes meet and the blocks were dark: violet S past the old blue/violet boundary at 128 (Table X) and orange S at H=4–6 past the old red/orange boundary at 6; we re-fitted both (to 119 and 2) from thousands of logged readings. The margin is still 1–2 hue units, so Events 2 and 3 also sort same-name blocks by hue rank. The violet still read as blue (H=114) shows the limit of hue alone: the dull violets had $S = 126$–142 and every true blue $S \geq 228$, so saturation would separate them. Both missed S cubes had ragged top faces with a fill just under the shape test's limit, which exists to reject arches and prisms, so lowering it would trade recall for distractors.

How the thresholds were set. We replayed the detector offline on saved RGB-D snapshots and, before any threshold change, compared the accepted blocks on every snapshot. A change was kept only if its losses were single frames of blocks at the board edge. A 1.15 aspect limit, for example, lost 16 real blocks and was reverted to 1.25, and a 20 mm height floor to drop thin pieces would have lost 58 real detections, so we added the level-top and ramp tests instead. Caching the tag depths fixed a block or hand on a tag tilting the table plane (a simulated 11-block board gave 5 to 12 detections; with the cache, 10 to 11 of 11). Event run logs, which record every rejected region and why, exposed the failures in Table X.

Known limitations: touching blocks merge into one region (flagged "2+ blocks?"), and a lying cylinder looks like two touching S blocks.

D. Pick-and-place

Because everything is planned before the arm moves, an unreachable target leaves the arm still, and the step queue lets STOP act between moves. Most failures in practice came from the board model, not IK: a block that slipped and was not seen until the next look, and an uncovered block gripped where it was assumed to be rather than where it was seen, which we fixed by looking again before gripping. The gripper still cannot sense a missed grasp.

TABLE X
DETECTOR FAILURES FOUND IN EVENT RUN LOGS AND THE FIXES.

Failure (event)	Cause in the log	Fix
17 mm thin piece stacked as an S (E1)	top 16.4–18.1 mm, on the old -8 mm limit	lower bound -6.8 mm; all 79 logged frames rejected
Violet S placed as blue (E3)	hue 123–125, past the old 128 boundary	blue/violet boundary 119
S-M-S stack rejected (E2)	solver assumed larger blocks at the bottom	stacks in any order

Event 2's lost points came from the line gaps. We left 12 mm (S) and 10 mm (M) between blocks so the detector would see them as separate, without knowing how strictly lines would be judged. Closing them would hurt detection (at 10 mm the finished M line already merged into "2+ blocks?" regions), so a tighter line would have to be placed from the board model alone or pushed together after the last look.

V. CONCLUSION

We built a complete perception-to-action pipeline for the Lite 6: verified FK, closed-form IK with a numerical cross-check, AprilTag calibration with depth registration and bias correction, and a height-based detector that rejects distractors. The weakest link is 3D accuracy: about 9 mm of range error along each ray and a nearly constant $+11$ mm offset in x, most likely from the hand-measured fixture offset; the per-pixel bias correction in `image_to_world` corrected the heights of clicked points. Future work: grasp sensing, frame-to-frame tracking, and calibration over many tag poses instead of four coplanar tags.

REFERENCES

[1] University of Michigan Robotics, "ROB 550 Armlab Documentation," https://rob550-docs.github.io/docs/armlab/, 2026.

[2] UFACTORY, "Kinematic and Dynamic Parameters of the UFACTORY Lite 6," https://docs.supportarticle.ufactory.cc/support_articles/developer/kinematic-and-dynamic-parameters/lite6.html, 2024.

[3] K. M. Lynch and F. C. Park, *Modern Robotics: Mechanics, Planning, and Control*. Cambridge University Press, 2017.

[4] M. W. Spong, S. Hutchinson, and M. Vidyasagar, *Robot Modeling and Control*. Wiley, 2006.

[5] D. W. Marquardt, "An algorithm for least-squares estimation of nonlinear parameters," *Journal of the Society for Industrial and Applied Mathematics*, vol. 11, no. 2, pp. 431–441, 1963.

[6] G. Bradski, "OpenCV: Open source computer vision library," https://opencv.org, 2000.
[7] Z. Zhang, "A flexible new technique for camera calibration," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 22, no. 11, pp. 1330–1334, 2000.
[8] J. Wang and E. Olson, "AprilTag 2: Efficient and robust fiducial detection," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2016, pp. 4193–4198.
[9] R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed. Cambridge University Press, 2003.

VI. Appendices

Videos

Teach-and-repeat (Task 1.4): https://youtube.com/shorts/VigqkL3qs3I
Calibration (Task 2.5): https://drive.google.com/file/d/1TbOvOygcjNLAFKHqSr41Q7FgqYZJu4LQ/view
Click-to-grab, workspace view: https://drive.google.com/file/d/12HTXf8ZbybxixmD622wZucksWGqnoUk7/view
Click-to-grab, screen recording: https://drive.google.com/file/d/1qWlriA_QvnAngmzrVxZkt5t8hwMGpJ24/view

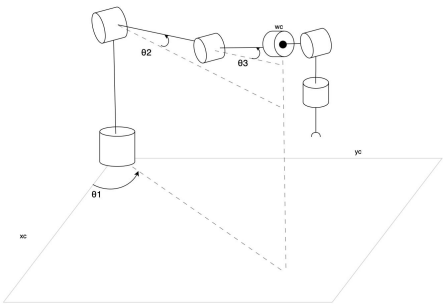


Fig. 13. IK schematic in 3D: wrist center (black dot) and the joint angles θ_1–θ_3.

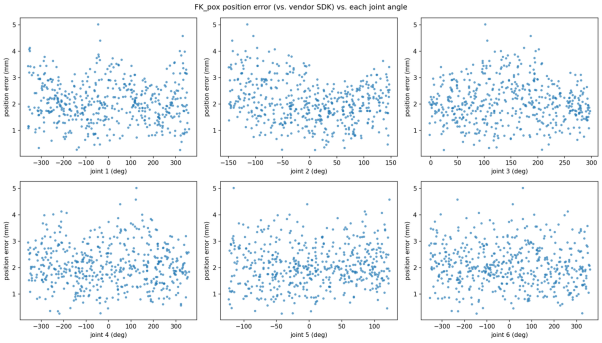


Fig. 11. PoX position error against the vendor vs. each joint angle: no structure.

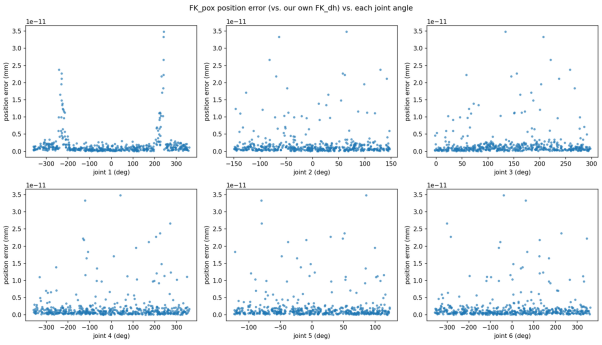


Fig. 12. PoX vs. DH position error vs. each joint angle (scale 10^{-11} mm).

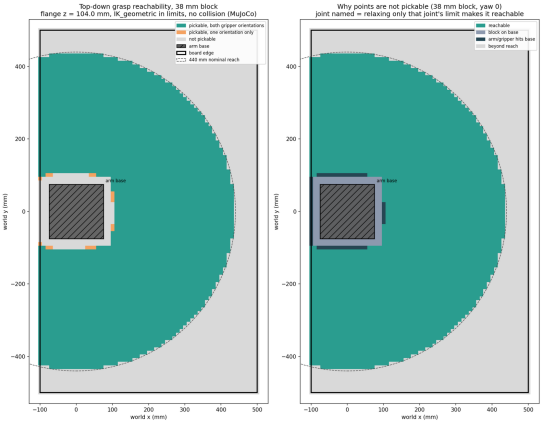


Fig. 14. Reachability for a 38 mm block, with the cause of each unreachable point.

TABLE XI
TAUGHT STACKING SEQUENCE (TASK 1.4). ANGLES IN RADIANS; GRIPPER 1 = CLOSED.

#	q_1	q_2	q_3	q_4	q_5	q_6	G
0	0.5037	0.5767	1.3765	0.0560	0.8509	0.5275	0
1	0.4832	1.0058	1.3682	0.0401	0.3809	0.3873	0
2	0.4832	1.0057	1.3681	0.0897	0.3809	0.3869	1
3	0.4832	0.3677	1.4211	0.0431	1.0830	0.4782	1
4	0.0333	0.7884	1.9203	0.0123	1.1882	0.0683	1
5	−0.0069	1.2727	1.8993	0.0738	0.6694	−0.0401	1
6	−0.0069	1.2727	1.8993	0.0771	0.6692	−0.0405	0
7	−0.0069	0.7569	1.9010	−0.1511	1.0896	−0.0483	0
8	−0.0073	0.2199	0.9463	−0.0384	0.7121	0.0255	0
9	−0.0050	0.9124	1.1610	−0.0817	0.2934	0.0332	0
10	−0.0050	0.9124	1.1610	−0.0811	0.2931	0.0324	1
11	−0.0049	0.3401	1.4081	0.0109	1.1088	0.0464	1
12	0.0328	0.8629	1.8397	−0.1116	0.9825	0.0008	1
13	−0.0037	1.1113	1.7745	0.0667	0.6387	−0.0451	1
14	−0.0037	1.1113	1.7745	0.0817	0.6387	−0.0464	0
15	−0.0038	0.5553	1.7794	0.0127	0.9607	−0.0589	0
16	−0.6206	0.3314	1.3067	0.0788	0.9277	−0.6705	0
17	−0.5049	1.0156	1.3736	0.0173	0.3710	−0.4888	0
18	−0.5049	1.0156	1.3737	0.0869	0.3710	−0.4901	1
19	−0.5050	0.4568	1.3985	0.0842	0.8172	−0.5145	1
20	0.0303	0.4624	1.7504	0.0038	1.2055	0.0140	1
21	0.0297	1.0523	1.8373	−0.0696	0.8228	0.1034	1
22	0.0297	1.0523	1.8373	−0.0748	0.8227	0.1018	0
23	0.0297	0.5485	1.8451	−0.1730	1.2158	0.0859	0
24	0.0000	0.1728	0.5550	0.0000	0.3822	0.0000	0

TABLE XII
TWELVE-POSE 3D CHECK (MM). TRUTH = SDK FLANGE + FIXTURE OFFSET; ESTIMATE = TAG VIA `IMAGE_TO_WORLD` (NO BIAS CORRECTION).

Loc	z	Truth	Estimate	e_x	e_y	e_z	$\|e\|$
A	low	(338.0, 1.1, 71.3)	(348.7, −0.2, 64.3)	10.7	−1.3	−7.0	12.9
A	mid	(338.0, 1.1, 171.6)	(346.6, −0.3, 164.0)	8.6	−1.4	−7.7	11.6
A	high	(337.5, 1.1, 271.0)	(344.5, −0.7, 261.8)	7.0	−1.7	−9.2	11.7
B	low	(122.8, −210.7, 71.3)	(128.8, −214.5, 54.8)	6.0	−3.8	−16.5	18.0
B	mid	(122.8, −210.7, 171.1)	(126.5, −216.7, 154.2)	3.7	−5.9	−16.8	18.2
B	high	(122.8, −210.7, 271.1)	(123.5, −218.6, 251.0)	0.7	−7.9	−20.0	21.5
C	low	(241.5, 240.2, 71.0)	(251.5, 242.8, 59.5)	10.0	2.6	−11.5	15.4
C	mid	(241.5, 240.2, 171.2)	(249.9, 242.2, 160.1)	8.4	2.0	−11.2	14.1
C	high	(241.5, 240.2, 271.1)	(247.2, 244.7, 258.4)	5.7	4.4	−12.6	14.5
D	low	(416.7, 256.0, 71.1)	(431.1, 256.1, 65.9)	14.3	0.1	−5.2	15.2
D	mid	(416.7, 256.0, 171.1)	(429.7, 256.8, 165.4)	12.9	0.8	−5.7	14.2
D	high	(416.7, 256.0, 271.4)	(428.0, 255.9, 265.4)	11.3	0.0	−5.9	12.7

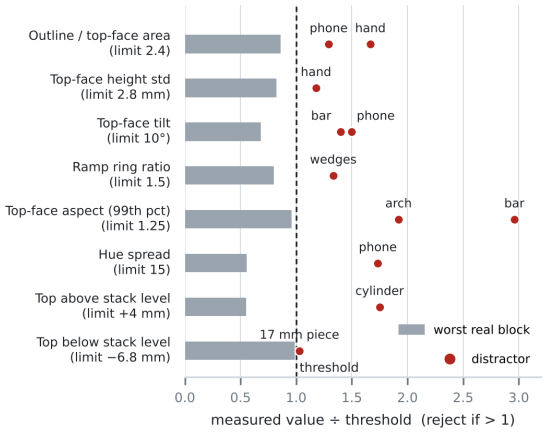


Fig. 16. Margin of each rejection test: real blocks (gray bar: worst value, or the percentile in Table III) and the distractors it rejects (red dots), as a fraction of the threshold. Right of the dashed line is rejected.

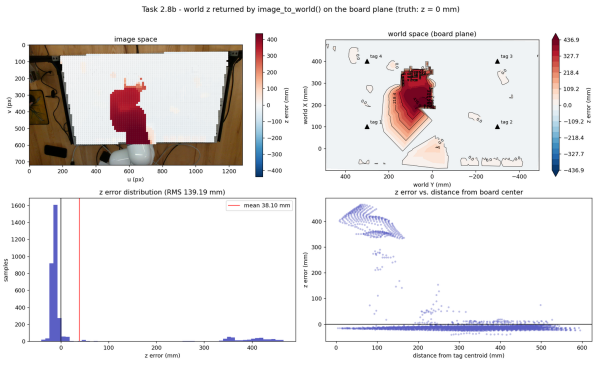


Fig. 15. Full board-plane error report (Task 2.8b), Oct. 15 session.